

NAIN, NL, Canada

WMO: 716650

Lat: 56.550N

Lon: 61.683W

Elev: 25

StdP: 14.68

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.8	-16.1	-31.1	1.0	-17.7	-27.5	1.2	-15.4	46.2	5.5	42.0	6.4	13.6	280	0.912

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.3	72.4	60.0	67.4	57.4	63.3	55.0	61.6	70.2	58.5	66.4	55.6	62.5	10.9	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.4	70.4	66.5	54.2	62.6	62.9	51.3	56.1	60.0	27.6	70.1	25.5	66.7	23.7	62.5	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 37.9	33.2	29.1	DB	-23.2	83.0	3.5	4.5	-25.7	86.3	-27.7	88.9	-29.7	91.4	-32.2	94.7	(4)
(5)			WB	-23.9	67.8	3.4	2.7	-26.4	69.8	-28.4	71.4	-30.3	72.9	-32.8	74.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	2.5	1.5	10.5	24.0	35.3	43.8	50.4	51.9	46.1	36.8	25.5	12.9
	DBStd	19.50	12.37	11.61	11.55	8.83	5.61	6.70	7.02	6.08	5.81	5.48	7.19	11.26
	HDD50	8077	1471	1358	1224	780	459	210	84	48	147	411	735	1150
	HDD65	13300	1936	1778	1689	1230	922	637	456	409	568	875	1185	1615
	CDD50	259	0	0	0	0	2	25	96	106	29	1	0	0
	CDD65	7	0	0	0	0	0	1	3	2	1	0	0	0
	CDH74	113	0	0	0	0	0	25	52	33	3	0	0	0
	CDH80	27	0	0	0	0	0	7	13	7	0	0	0	0
Wind	WSAvg	11.7	13.8	12.9	13.1	12.4	9.8	9.2	8.3	8.3	11.2	12.8	14.1	15.0
Precipitation	PrecAvg	35.6	3.4	2.4	2.9	2.8	2.2	2.9	4.2	3.0	3.0	3.0	2.6	2.7
	PrecMax	47.8	9.9	6.1	6.9	6.8	5.8	4.8	7.5	7.6	7.2	6.4	4.6	6.0
	PrecMin	25.7	0.4	0.0	0.5	0.4	0.1	0.7	1.1	1.1	0.7	1.5	0.6	0.6
	PrecStd	5.0	2.1	1.6	1.9	1.6	1.3	1.1	1.8	1.4	1.7	1.0	1.0	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.5	35.5	40.3	46.4	62.6	77.3	80.5	78.1	71.4	57.4	43.8	36.9
		MCWB	34.4	34.6	35.9	39.6	51.1	61.2	61.9	62.4	59.4	49.9	40.2	35.7
	2%	DB	33.1	32.2	35.3	42.3	53.2	65.8	72.0	71.7	64.3	51.0	39.5	34.3
		MCWB	32.1	31.1	32.9	37.0	45.0	54.7	59.1	60.5	56.6	45.2	36.7	33.3
	5%	DB	29.2	26.0	32.3	38.9	47.1	59.7	67.1	66.6	59.4	47.0	37.2	32.5
		MCWB	28.0	24.4	30.7	34.9	41.1	51.1	57.2	57.6	52.5	42.2	35.0	31.5
	10%	DB	23.0	19.8	28.6	35.8	43.2	54.4	62.6	62.6	55.6	43.9	35.0	29.5
		MCWB	21.6	18.2	26.6	33.1	38.4	47.1	54.4	55.2	49.0	39.7	33.0	28.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.8	34.9	36.4	40.5	51.5	63.5	64.4	65.9	61.7	50.6	41.1	35.9
		MCDB	35.3	35.4	39.3	45.5	62.5	73.2	75.8	76.0	68.5	56.0	43.1	36.8
	2%	WB	32.2	31.3	33.4	37.5	45.6	56.2	60.9	61.9	57.2	45.9	37.3	33.6
		MCDB	33.1	32.1	35.0	41.7	52.4	64.3	70.4	69.4	63.9	50.2	38.6	34.2
	5%	WB	28.0	24.5	30.7	35.4	41.4	51.4	57.8	58.5	53.0	42.6	35.2	31.7
		MCDB	29.0	25.9	32.5	38.2	46.7	59.2	66.1	65.9	59.3	46.4	36.9	32.6
	10%	WB	21.6	18.3	26.7	33.6	38.6	47.4	54.9	55.6	49.5	40.5	33.3	28.2
		MCDB	23.0	19.8	28.5	35.5	42.7	53.8	62.0	62.0	54.7	43.2	34.8	29.5
Mean Daily Temperature Range	5% DB	MDBR	12.3	13.5	14.9	13.6	11.9	13.9	14.1	14.3	12.3	8.6	8.8	10.2
		MCDBR	14.1	14.6	16.5	16.6	20.3	24.8	24.5	23.0	19.6	13.7	10.1	8.5
		MCWBR	14.2	14.0	15.3	13.7	14.3	16.1	15.2	14.9	13.9	10.4	9.4	8.6
	5% WB	MCDBR	14.2	14.5	15.8	14.3	19.2	23.8	23.4	22.6	18.9	12.7	9.5	7.8
		MCWBR	14.4	14.0	15.0	12.2	13.9	16.6	15.6	15.6	14.3	10.1	9.3	8.1
Clear-Sky Solar Irradiance	taub	0.230	0.250	0.277	0.331	0.398	0.379	0.383	0.364	0.331	0.294	0.250	0.235	
	taud	2.212	2.246	2.199	2.094	2.063	2.332	2.400	2.458	2.513	2.529	2.298	2.104	
	Ebn at Noon	220	263	282	279	265	273	270	268	263	247	209	175	
	Edh at Noon	16	24	33	43	48	37	34	30	25	19	15	14	
All-Sky Solar Radiation	RadAvg	204	521	1000	1510	1815	1663	1454	1252	903	469	219	137	
	RadStd	29	64	114	135	171	178	135	119	67	34	22	15	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page